

TopoLite-KD: Efficient Topology-Aware Knowledge Distillation for COVID-19 CT Slice Classification

Ibadat Ali, Shawaiz Ali, Muhammad Abdullah, and Muhammad Usama
Department of Computer Science, GIFT University, Gujranwala, Pakistan

Abstract—Accurate COVID-19 computed-tomography (CT) classification is often pursued with computationally expensive ensembles, while lightweight models may discard structural information that is not readily captured by local appearance features. We formulate slice-level COVID-19 recognition as supervised binary classification and present TopoLite-KD, a 196,773-parameter student that combines a depthwise-separable visual encoder, a fixed 134-dimensional persistent-homology descriptor, direct input-conditioned visual-topological gated fusion, and supervised response-based distillation from an EfficientNet-B0 teacher. The topology descriptor is deterministic and label-free, but the end-to-end system does not use unsupervised or self-supervised representation learning. On a duplicate-audited image-level 70/15/15 split of the SARS-CoV-2 CT-scan dataset, the verified seed-42 model achieved 94.62% accuracy (95% bootstrap confidence interval: 92.47–96.77%), 96.81% sensitivity, 92.39% specificity, 0.8932 Matthews correlation coefficient, and 0.9846 AUROC. Five-seed ablations identify topology as the dominant signal, while also showing that the complete gated, distilled configuration is not the strongest tested variant: simple concatenation and removal of Coordinate Attention reached 95.16% and 95.27% mean accuracy, respectively, versus 94.41% for the complete model. We therefore claim a compact and auditable integration rather than state-of-the-art accuracy, and explicitly separate verified measurements from non-reproducible archival records. Code is available in the [TopoLite-KD GitHub repository](#).

Index Terms—COVID-19, computed tomography, persistent homology, topological data analysis, knowledge distillation, lightweight neural networks, medical image classification

I. INTRODUCTION

Automated analysis of chest CT images was widely investigated as a research aid during the COVID-19 pandemic. Public datasets enabled rapid model development, but also encouraged comparisons across incompatible data splits, preprocessing pipelines, and validation procedures. Reviews of COVID-19 imaging models have warned that image-level leakage, inadequate external validation, and optimistic model selection can produce results that do not generalize clinically [4]. Consequently, accuracy alone is insufficient: a credible method should also expose its data provenance, evaluation protocol, computational footprint, and failure modes.

The assigned reference method of Kundu *et al.* combines VGG-11, GoogLeNet, SqueezeNet v1.1, and Wide ResNet-50-2 using a Sugeno fuzzy integral and reports 98.93% accuracy and sensitivity [2]. Its central strength is adaptive ensemble fusion. Its central deployment limitation is structural: each prediction requires four pretrained convolutional networks. Moreover, its public implementation and README do not

define a protocol directly comparable with our frozen 70/15/15 split; the README requests an 80/20 train/validation split, whereas the ensemble script hard-codes 745 validation labels and reuses the validation partition for model selection and probability extraction [3]. We therefore treat 98.93% as a published literature result, not as a reproduced control.

Our motivation is to investigate whether a single compact student can combine three complementary sources of inductive bias: (i) learned image appearance, (ii) explicit multithreshold topology, and (iii) soft supervision from a stronger teacher. Persistent homology summarizes connected components and loops across an intensity filtration, offering information that is not restricted to local convolutional responses [5]. Knowledge distillation transfers softened teacher responses to a smaller student [8], while EfficientNet provides a well-established accuracy–efficiency teacher family [9].

This work makes the following evidence-bounded contributions:

- 1) We implement a 196,773-parameter dual-branch student that integrates a custom depthwise-separable CNN with a fixed 134-D persistent-homology descriptor.
- 2) We formulate direct reliability-aware, per-feature gated fusion of aligned 64-dimensional visual and topological embeddings, and train the student with EfficientNet-B0 response distillation and two branch-supervision heads.
- 3) We provide a leakage-audited frozen split, validation-only calibration and threshold selection, per-sample predictions, bootstrap confidence intervals, profiling, and five-seed ablations for eight completed configurations.
- 4) We separate reproduced measurements from archival author records and show that the available ablations support the value of topology, but not the superiority of every proposed component.

Accordingly, the primary contribution is the specific compact integration and its auditable evaluation, rather than the invention of persistent homology, EfficientNet, Coordinate Attention, or knowledge distillation individually.

II. RELATED WORK

A. COVID-19 CT Classification

The SARS-CoV-2 CT-scan dataset was introduced as a public collection of positive and non-COVID CT images from hospitals in São Paulo, Brazil [1]. Kundu *et al.* used four pretrained CNNs and a Sugeno fuzzy integral to adaptively aggregate class scores [2]. The reported 98.93% result demonstrates the predictive capacity of a multi-backbone ensemble,

but it does not answer whether comparable discrimination can be approached with a sub-million-parameter student.

Protocol sensitivity is particularly important in this dataset family. The dataset contains multiple slices associated with a limited number of patients, while our supplied file tree does not expose reliable patient identifiers. A random image-level split can therefore place correlated slices across partitions. We retain the measured image-level result for reproducibility, but do not interpret it as patient-level or external clinical validation.

B. Topology in Medical Imaging

Topological data analysis (TDA) represents structures through properties that persist across a filtration. In images, zero-dimensional homology (H_0) represents connected components and one-dimensional homology (H_1) represents loops or enclosed regions. Singh *et al.* review the growing role of persistent homology in medical imaging [5]. Iqbal *et al.* applied homological features to the same COVID-19 CT dataset [6], and Assaf *et al.* later combined persistent-homology features with conventional machine learning for COVID-19 CT classification [7]. These studies motivate topology as an informative representation, but neither establishes the behavior of topology inside the lightweight teacher–student fusion system studied here.

C. Efficient Networks and Knowledge Distillation

Knowledge distillation trains a compact student to match temperature-softened outputs from a teacher [8]. EfficientNet scales depth, width, and resolution jointly, yielding strong transferable CNNs at controlled computational cost [9]. Distillation has since been used broadly in medical imaging, including classification, segmentation, and limited-data learning [10]. Tan *et al.* combined self-supervised learning and self-distillation for COVID-19 image classification [11]; however, self-distillation and our frozen external-teacher response distillation are different training paradigms.

The student also uses Group Normalization, which is independent of batch statistics [12], and Coordinate Attention, which factorizes spatial aggregation along height and width [13]. Importantly, these modules are established prior work. Our novelty claim is limited to their evaluated integration with the fixed topology descriptor and distillation objective.

III. ARCHITECTURE

Figure 1 separates inference-time components from training-only supervision. The deployed student retains the visual encoder, topology encoder, reliability-aware fusion module, and binary classifier. The EfficientNet-B0 teacher and the two branch-supervision heads affect optimization only and are not required during student inference.

IV. METHODOLOGY

A. Learning Paradigm

TopoLite-KD is a supervised teacher–student classifier. Ground-truth binary labels supervise the main and branch-level

binary-cross-entropy terms, while probabilities from a separately label-trained EfficientNet-B0 teacher provide response-based distillation targets. The persistent-homology branch performs deterministic, label-free feature extraction; it is not an unsupervised learned representation. No self-supervised pre-training, pseudo-labeling, clustering objective, or unlabeled-data loss is used in the reported student experiments. This distinction separates our setting from self-supervised self-distillation approaches [11].

B. Problem Formulation

Let x denote one grayscale CT slice and let $y \in \{0, 1\}$ denote its binary label, where $y = 1$ represents COVID-19 and $y = 0$ represents Non-COVID-19. Two preprocessing functions generate the inputs required by the visual and topology branches:

$$x_v = \mathcal{P}_v(x) \in \mathbb{R}^{1 \times 224 \times 224}, \quad (1)$$

$$x_t = \mathcal{P}_t(x) \in [0, 1]^{1 \times 64 \times 64}. \quad (2)$$

The two encoders produce aligned 64-dimensional embeddings,

$$e_v = E_v(x_v) \in \mathbb{R}^{64}, \quad e_t = E_t(\hat{d}_{\text{top}}) \in \mathbb{R}^{64}, \quad (3)$$

where $\hat{d}_{\text{top}}$ is the standardized 134-dimensional persistent-homology descriptor. The fused classifier produces a scalar logit a_f and probability

$$p = \text{sigmoid}(a_f) \in [0, 1]. \quad (4)$$

C. Lightweight Visual Encoder

The visual stem is a 3×3 convolution with stride two, Group Normalization, and SiLU activation, mapping one channel to 24 channels. Four stages then use two depthwise-separable residual blocks each. The first block in every stage down-samples and changes width; the second preserves resolution. Channel progression is

$$1 \rightarrow 24 \rightarrow 32 \rightarrow 48 \rightarrow 96 \rightarrow 160. \quad (5)$$

Coordinate Attention is applied after the 32- and 48-channel stages in the full configuration. Global average pooling converts the final 160-channel feature map into a compact vector. A two-layer projection then produces the 64-dimensional visual embedding

$$e_v = \text{LN}(W_2 \text{Drop}[\text{SiLU}(\text{LN}(W_1 h))]) \in \mathbb{R}^{64}, \quad (6)$$

where h is the globally pooled 160-dimensional feature vector and the hidden projection width is 192.

D. Persistent-Homology Descriptor

The unaugmented grayscale slice is resized to 64×64 and scaled to $[0, 1]$. Let I denote this intensity field. Two cubical filtrations are used:

$$F_{\text{sub}} = I, \quad F_{\text{sup}} = 1 - I. \quad (7)$$

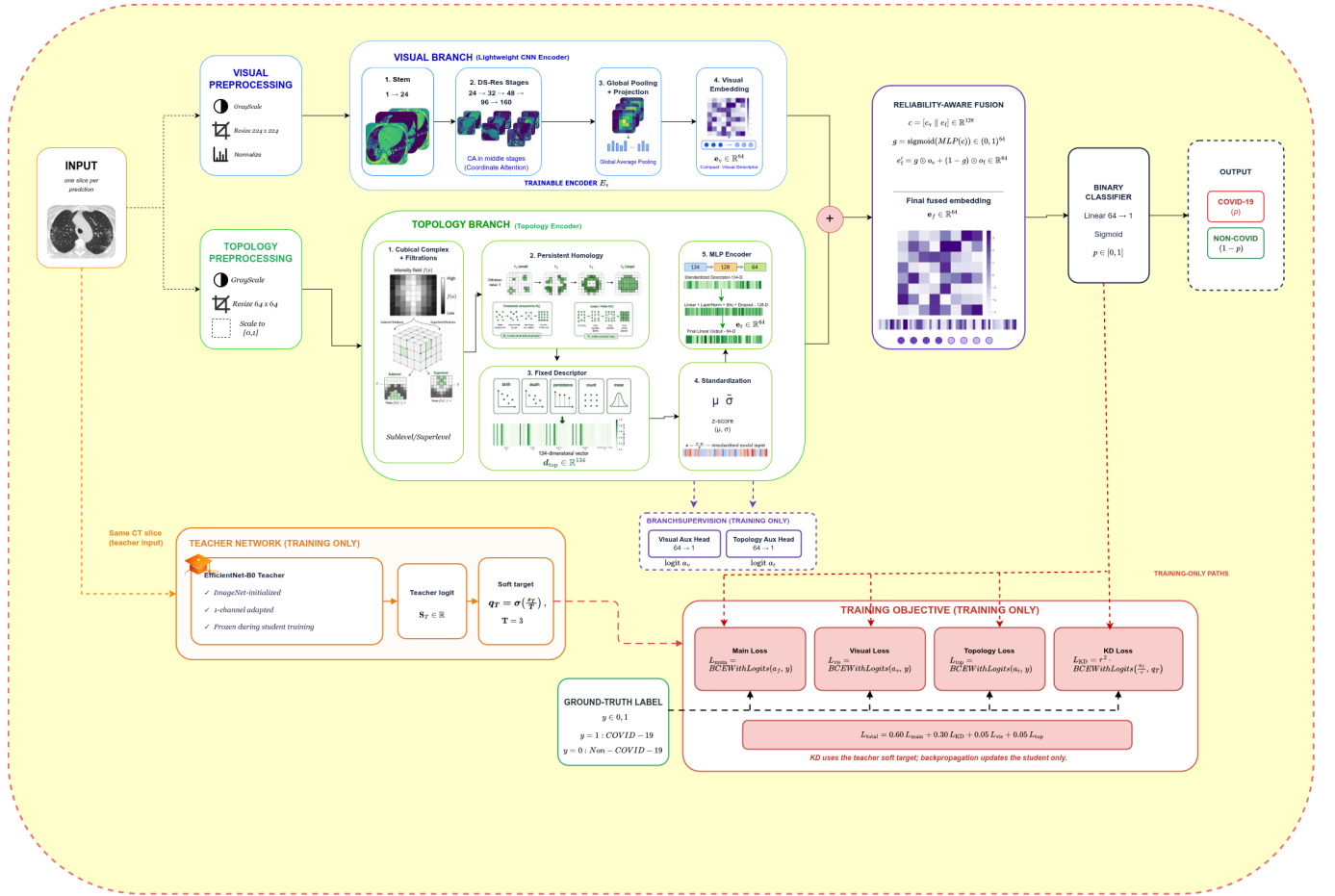


Fig. 1. Overall architecture of TopoLite-KD. A grayscale CT slice is processed through two complementary routes: a trainable lightweight visual encoder operating on a normalized 224×224 input and a deterministic topology encoder operating on an unaugmented 64×64 intensity field. Both branches produce 64-dimensional embeddings. Their concatenation parameterizes an input-conditioned 64-dimensional reliability gate that directly blends the visual and topological representations before a linear binary classifier. During training, a frozen EfficientNet-B0 teacher supplies temperature-softened targets, while visual and topology auxiliary heads provide branch-level supervision. Dashed connections denote training-only paths that are removed during inference.

For filtration $f \in \{\text{sub}, \text{sup}\}$ and homology dimension $k \in \{0, 1\}$, persistent homology produces intervals

$$D_{f,k} = \{(b_i, d_i)\}_{i=1}^{n_{f,k}}, \quad \ell_i = \max(d_i - b_i, 0), \quad (8)$$

where infinite deaths are clipped to the maximum filtration value. Each of the four (f, k) groups yields a 30-D block comprising ten lifetime statistics, four normalized counts above persistence thresholds $\{0.01, 0.03, 0.05, 0.10\}$, and a 16-point Betti curve. At filtration value u , the Betti feature is

$$\beta_{f,k}(u) = \frac{1}{256} \sum_i \mathbf{1}[b_i \leq u < d_i], \quad (9)$$

clipped to $[0, 1]$.

Seven binary thresholds $\{0.2, 0.3, \dots, 0.8\}$ provide seven connected-component counts and seven hole counts. Thus,

$$d_{\text{TDA}} = 4(10 + 4 + 16) + 14 = 134. \quad (10)$$

Descriptor mean μ and standard deviation σ are estimated on the training split only, and every descriptor is standardized as

$$\hat{d} = (d - \mu) \oslash (\sigma + \epsilon). \quad (11)$$

An MLP with dimensions $134 \rightarrow 128 \rightarrow 64$, Layer Normalization, SiLU, and dropout maps the standardized descriptor to

$$e_t = E_t(\hat{d}_{\text{top}}) \in \mathbb{R}^{64}. \quad (12)$$

E. Reliability-Aware Fusion

Because both branches already produce aligned 64-dimensional embeddings, the revised fusion module does not introduce separate visual and topology projection layers. The two embeddings are first concatenated:

$$c = [e_v \parallel e_t] \in \mathbb{R}^{128}. \quad (13)$$

An input-conditioned, per-feature gate is then computed as

$$g = \text{sigmoid}(\text{MLP}_g(c)) \in (0, 1)^{64}. \quad (14)$$

The final fused representation is the direct gated blend

$$e'_f = g \odot e_v + (1 - g) \odot e_t \in \mathbb{R}^{64}, \quad (15)$$

where $\odot$ denotes element-wise multiplication. A gate value near one emphasizes the corresponding visual feature, whereas

a value near zero emphasizes the corresponding topological feature. The main binary classifier is

$$a_f = w_f^\top e'_f + b_f, \quad p = \text{sigmoid}(a_f). \quad (16)$$

Two training-only linear heads provide branch supervision:

$$a_v = w_v^\top e_v + b_v, \quad a_t = w_t^\top e_t + b_t. \quad (17)$$

F. EfficientNet-B0 Teacher and Training Objective

The ImageNet-initialized EfficientNet-B0 teacher is adapted to grayscale by replacing its first convolution with a one-channel kernel initialized by averaging the pretrained RGB kernels. Its classifier is replaced by a single-logit binary head. The teacher is fine-tuned on the same frozen training partition, selected by validation MCC, and then frozen during student training.

Let $s_T \in \mathbb{R}$ denote the frozen teacher logit. With distillation temperature $T = 3$, the teacher soft target is

$$q_T = \text{sigmoid}\left(\frac{s_T}{T}\right), \quad T = 3. \quad (18)$$

The main supervised loss is

$$\mathcal{L}_{\text{main}} = \text{BCEWithLogits}(a_f, y). \quad (19)$$

The branch-supervision losses are

$$\mathcal{L}_{\text{vis}} = \text{BCEWithLogits}(a_v, y), \quad (20)$$

$$\mathcal{L}_{\text{top}} = \text{BCEWithLogits}(a_t, y). \quad (21)$$

Binary response distillation is applied between the temperature-scaled fused student logit and the teacher target:

$$\mathcal{L}_{\text{KD}} = T^2 \text{BCEWithLogits}\left(\frac{a_f}{T}, q_T\right). \quad (22)$$

The complete student-training objective is

$$\begin{aligned} \mathcal{L}_{\text{total}} = & 0.60 \mathcal{L}_{\text{main}} + 0.30 \mathcal{L}_{\text{KD}} \\ & + 0.05 \mathcal{L}_{\text{vis}} + 0.05 \mathcal{L}_{\text{top}}. \end{aligned} \quad (23)$$

The teacher target is treated as fixed supervisory information. Backpropagation updates only the student visual encoder, topology MLP, reliability gate, main classifier, and auxiliary heads.

G. Validation-Only Calibration and Decision Rule

After model selection, temperature scaling fits a scalar $\tau > 0$ on validation logits by minimizing validation binary cross-entropy [14]. For sample i , the calibrated probability is

$$p_i^{\text{cal}} = \text{sigmoid}\left(\frac{a_{f,i}}{\tau}\right). \quad (24)$$

The decision threshold is selected on validation predictions by maximizing Youden's index,

$$J(\theta) = \text{Sensitivity}(\theta) + \text{Specificity}(\theta) - 1. \quad (25)$$

The fitted τ and θ are fixed before test inference.

Algorithm 1: Training and frozen-test evaluation of TopoLite-KD

- 1) Fit the TDA standardizer (μ, σ) on training descriptors only.
- 2) Fine-tune teacher f_T on training data, select its checkpoint using validation MCC, and freeze it.
- 3) For each student minibatch (x, d, y) , compute $(e_v, e_t, e'_f, a_v, a_t, a_f) \leftarrow f_S(x, d)$ and $s_T \leftarrow f_T(x)$ without teacher gradients.
- 4) Update f_S using the weighted main, distillation, visual, and topology losses; select its checkpoint using validation MCC.
- 5) Fit temperature τ and threshold θ using validation predictions only.
- 6) Evaluate once on the frozen test set and save predictions, metrics, 2,000-sample bootstrap intervals, and audit hashes.

TABLE I
DUPLICATE-AUDITED FROZEN SPLIT USED BY THE VERIFIED RECONSTRUCTION.

Partition	COVID-19	Non-COVID	Total
Train	874	861	1,735
Validation	188	184	372
Test	188	184	372
Total	1,250	1,229	2,479

V. EXPERIMENTAL SETTINGS

A. Dataset and Split Audit

The executable notebook used the Kaggle copy of the SARS-CoV-2 CT-scan dataset [1]. The local audit discovered 2,481 files, removed two exact same-label duplicates by SHA-256, and retained 2,479 images. Table I summarizes the frozen split. No patient identifier could be extracted; therefore, the evaluation is explicitly image-level.

B. Training Configuration

Table II records the resolved configuration of the audited run. The visual branch receives augmented 224×224 inputs, whereas TDA is computed from the unaugmented 64×64 grayscale slice. This separation prevents random visual augmentation from changing the cached topological descriptor.

The Kaggle environment exposed two Tesla T4 GPUs, although the scripts selected a single CUDA device and did not use distributed training. The audited environment used Python 3.12.13, PyTorch 2.10.0+cu128, CUDA 12.8, and cuDNN 9.1. The main student stopped after epoch 42 because validation MCC had not improved for 12 epochs.

TABLE II
RESOLVED TRAINING AND EVALUATION CONFIGURATION.

Item	Setting
Student optimizer	AdamW, $\eta = 3 \times 10^{-4}$, weight decay 10^{-4}
Schedule	Cosine decay, 3 warm-up epochs, minimum LR 5×10^{-6}
Student budget	50 epochs; patience 12; validation MCC selection
Batch / precision	32; automatic mixed precision; gradient norm 5
Augmentation	Horizontal flip 0.5; rotation $\pm 10^\circ$; translation 0.03; scale 0.95–1.05; brightness/contrast 0.08
Normalization	Mean 0.5, standard deviation 0.5
Teacher	ImageNet EfficientNet-B0; 30-epoch cap; AdamW LR 10^{-4} ; patience 8
Distillation	Response KD, $T = 3$, weight 0.30
Calibration	Validation temperature scaling and Youden threshold
Uncertainty	2,000 bootstrap samples; 95% percentile interval
Ablation seeds	42, 1337, 2026, 3407, 9001

C. Evaluation Metrics

For true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), we report

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (26)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}, \quad (27)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (28)$$

$$F_1 = \frac{2 \text{ Precision Sensitivity}}{\text{Precision} + \text{Sensitivity}}, \quad (29)$$

$$\text{MCC} = \frac{TPTN - FPFN}{\Delta}, \quad (30)$$

$$\Delta = [(TP + FP)(TP + FN) \cdot (TN + FP)(TN + FN)]^{1/2}. \quad (31)$$

Threshold-independent metrics are AUROC and area under the precision–recall curve (AUPRC). Calibration is summarized using negative log-likelihood (NLL), Brier score, and 15-bin expected calibration error (ECE).

VI. RESULTS AND ANALYSIS

A. Quantitative Evaluation

The audited seed-42 run produced $TP = 182$, $TN = 170$, $FP = 14$, and $FN = 6$ at a validation-selected threshold of 0.4807. The fitted calibration temperature was 2.3506. Table III reports the test results and bootstrap intervals.

B. Efficiency Profile

The student has 196,773 trainable parameters, corresponding to 0.751 MiB of FP32 parameter storage. On one Tesla T4 with batch size one, 20 warm-up passes, and 100 timed iterations, the mean neural forward-pass latency was 4.05 ms, the median was 4.00 ms, and the 95th percentile was 4.41 ms.

TABLE III
VERIFIED TEST PERFORMANCE ON 372 FROZEN TEST IMAGES.

Metric	Point estimate	95% bootstrap CI
Accuracy (%)	94.62	92.47–96.77
Balanced accuracy (%)	94.60	92.42–96.76
Sensitivity (%)	96.81	94.15–98.94
Specificity (%)	92.39	88.59–96.20
Precision (%)	92.86	89.60–96.24
F_1 (%)	94.79	92.71–96.84
MCC	0.8932	0.8502–0.9357
AUROC	0.9846	0.9738–0.9932
AUPRC	0.9851	0.9747–0.9934
Brier score	0.0442	0.0286–0.0612
ECE	0.0349	0.0281–0.0604

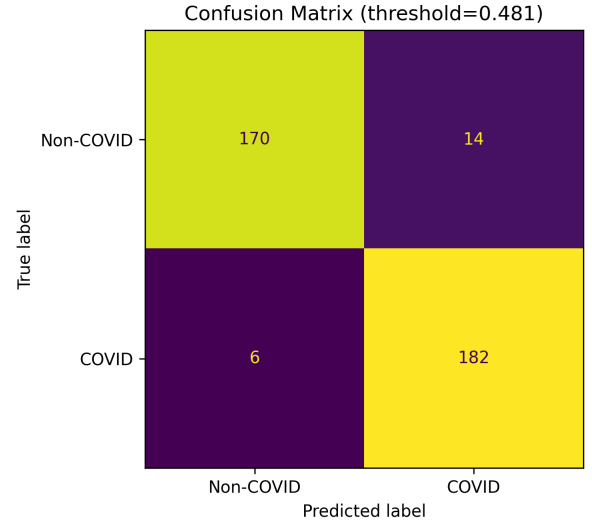


Fig. 2. Confusion matrix for the verified seed-42 TopoLite-KD model on 372 frozen test images at the validation-selected threshold $\theta = 0.4807$: $TP = 182$, $TN = 170$, $FP = 14$, and $FN = 6$.

Measured throughput was 247.2 images/s and peak allocated CUDA memory was 11.77 MiB. These measurements use a precomputed 134-D descriptor and therefore exclude image loading and CPU/GUDHI persistent-homology extraction; they are not end-to-end clinical latency.

C. Five-Seed Component Ablation

Table IV summarizes all 40 completed ablation runs. Values are mean $\pm$ sample standard deviation over five seeds on the same 372-image frozen test split.

Three conclusions follow. First, the TDA-only branch improves mean accuracy by 5.59 percentage points over the visual-only model, indicating that the fixed topological descriptor is the dominant source of performance in this reconstruction. Second, simple concatenation is the strongest completed fusion configuration by AUROC and shows lower observed seed-to-seed variability than several alternatives, although no formal paired test is available. Third, neither KD nor Coordinate Attention receives positive support from these ablations: visual+KD is 0.38 points below visual-only,

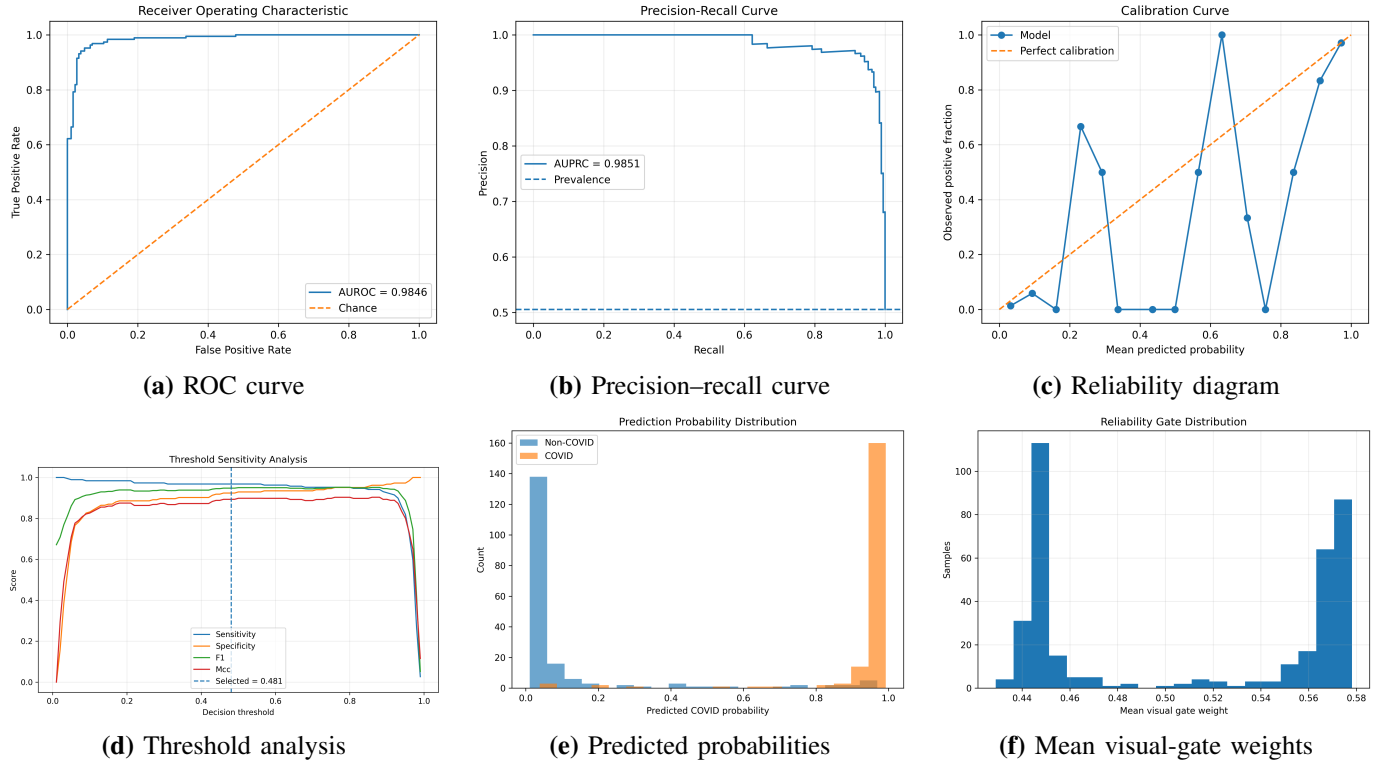


Fig. 3. Quantitative diagnostics for the verified A6 seed-42 model: (a) AUROC = 0.9846; (b) AUPRC = 0.9851; (c) test reliability after validation-fitted temperature scaling; (d) threshold sensitivity, with the dashed line marking the validation-selected $\theta = 0.4807$; (e) class-conditional COVID-19 probabilities; and (f) sample-level mean visual-gate weights. The gate histogram is descriptive and does not establish calibrated modality reliability.

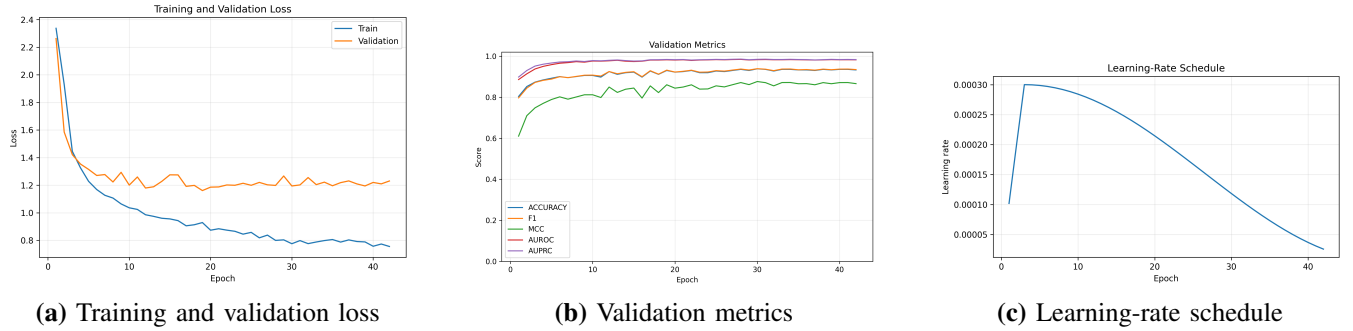


Fig. 4. Training diagnostics for the verified seed-42 run: (a) optimization curves, (b) validation accuracy, F_1 , MCC, AUROC, and AUPRC, and (c) warm-up followed by cosine learning-rate decay. Checkpoint selection used validation MCC rather than test performance.

TABLE IV
FIVE-SEED ABLATION RESULTS. THE COMPLETE MODEL IS A6. BOLD INDICATES THE BEST MEAN IN A COLUMN.

ID	Configuration	Accuracy (%)	Sensitivity (%)	Specificity (%)	MCC	AUROC (%)
A0	Visual only	89.30 $\pm$ 0.64	86.70 $\pm$ 1.64	91.96 $\pm$ 1.24	0.7875 $\pm$ 0.0123	94.97 $\pm$ 0.29
A1	TDA only	94.89 $\pm$ 0.66	96.60 $\pm$ 0.81	93.15 $\pm$ 1.47	0.8984 $\pm$ 0.0130	98.69 $\pm$ 0.15
A2	Visual + TDA concatenation	95.16 $\pm$ 0.43	96.70 $\pm$ 1.02	93.59 $\pm$ 0.71	0.9037 $\pm$ 0.0086	98.72 $\pm$ 0.13
A3	Fixed 0.5/0.5 fusion	94.30 $\pm$ 0.58	97.45 $\pm$ 1.02	91.09 $\pm$ 0.62	0.8877 $\pm$ 0.0120	98.69 $\pm$ 0.19
A4	Gated fusion, no KD	94.25 $\pm$ 0.52	97.13 $\pm$ 0.71	91.30 $\pm$ 0.77	0.8863 $\pm$ 0.0105	98.51 $\pm$ 0.19
A5	Visual + KD, no TDA	88.92 $\pm$ 1.90	87.45 $\pm$ 2.73	90.43 $\pm$ 3.56	0.7797 $\pm$ 0.0376	94.77 $\pm$ 1.28
A6	Full TopoLite-KD	94.41 $\pm$ 0.22	96.81 $\pm$ 1.25	91.96 $\pm$ 1.05	0.8893 $\pm$ 0.0048	98.46 $\pm$ 0.11
A7	Full model without CoordAtt	95.27 $\pm$ 0.67	97.34 $\pm$ 0.38	93.15 $\pm$ 1.19	0.9061 $\pm$ 0.0131	98.41 $\pm$ 0.20

and removing Coordinate Attention raises mean accuracy by 0.86 points relative to A6. A8 (H_0 only), A9 (H_1 only), and A10 (single-filtration) configurations exist in the code package

but are omitted from Table IV because completed checkpoint-backed metric artifacts were not found in the audited A0–A7 recovery suite. A separately supplied console log describes

additional A8–A10 runs under different seeds; those log-only results are not merged into this artifact-backed table.

D. Qualitative Evaluation

Grad-CAM was used to inspect the visual branch qualitatively [15]. Figure 5 presents two correctly classified Non-COVID and two correctly classified COVID-19 slices. These maps show regions that influenced the visual prediction, but they are not lesion segmentations and were not validated by a radiologist.

E. Evidence Provenance and Baseline Comparison

Table V prevents three different result types from being conflated. The published fuzzy-integral result is not a rerun under our split. The 96.77% TopoLite-KD value is an archival author record. Only the 94.62% row has a matching resolved configuration, checkpoint, predictions, metric files, bootstrap intervals, logs, environment record, and artifact manifest in the supplied audited package.

The 4.31-point gap between the published ensemble and the audited reconstruction cannot be attributed solely to architecture. The methods differ in split construction, duplicate handling, model-selection data, thresholding, calibration, and possibly dataset version. Therefore, we do not claim that TopoLite-KD matches or statistically underperforms the baseline under a controlled experiment; we state only that the published headline is higher.

VII. DISCUSSION

A. What the Results Support

The experiments support a narrow but useful conclusion: explicit topology can provide a strong, compact representation of this dataset. A 134-D descriptor processed by a small MLP outperforms the visual branch by a large margin, and concatenating the two modalities yields the strongest AUROC among completed configurations. This aligns with prior evidence that persistent homology can summarize medically relevant structures across intensity thresholds [6], [7].

The results do not support stronger claims that every component is beneficial or that the full model is state of the art. The gated fusion does not surpass concatenation, response KD does not improve the visual-only student, and Coordinate Attention decreases mean accuracy in the controlled ablation. These negative results are scientifically informative. They suggest that the topology branch may already encode most separable information in this split, while the teacher signal and learned gate may introduce optimization conflict or redundant evidence. Additional tuning could change these outcomes, but such improvement is a hypothesis, not a result.

B. Contribution Relative to Prior Work

The method is inspired by three established lines of research: compact depthwise-separable networks, persistent homology, and teacher–student learning. Its contribution is their concrete integration under a direct per-feature reliability gate, the fixed-width descriptor that preserves model capacity across

topology ablations, and an auditable evaluation package. We deliberately avoid a “first” claim because the literature already contains TDA-based COVID-19 CT classification [6], [7] and distillation-based COVID-19 image classification [11].

Compared with the assigned four-CNN fuzzy ensemble, TopoLite-KD replaces multi-backbone inference with one small student and precomputed topological features. The resulting parameter count is independently measured, but a compression ratio is not reported because the baseline’s exact deployed parameter footprint was not supplied or reprofiled in a matched environment.

C. Limitations

This study has several consequential limitations:

- 1) The split is image-level, not patient-level. Correlated slices from the same patient may cross partitions because patient identifiers were unavailable.
- 2) Evaluation uses one public dataset and no external institution, scanner, geographic cohort, or prospective study. The system is research software and must not be interpreted as a clinical diagnostic device.
- 3) The baseline result was not reproduced under the same manifest. Its public README and scripts also contain split and evaluation inconsistencies, preventing a clean controlled comparison.
- 4) The historical 96.77% record lacks its exact manifest, checkpoint, raw predictions, and environment, so it cannot anchor the principal claim.
- 5) A8–A10 topology ablations are incomplete, and the available 40-run ablation suite has no paired significance-test output.
- 6) Reported latency excludes persistent-homology extraction, data transfer, and image decoding. End-to-end latency and energy consumption remain to be measured.
- 7) Grad-CAM visualizations were generated [15], but no radiologist localization study or explanation-faithfulness analysis was conducted.

Future work should begin with patient-grouped resplitting using verified patient identifiers, external validation, and a matched rerun of the fuzzy-integral baseline. Methodologically, the strongest immediate baseline is A2 concatenation, not A6. Future gated or distillation variants should therefore be required to beat A2 across multiple patient-level seeds with paired confidence intervals before being promoted as improvements.

VIII. CONCLUSION

We presented an evidence-audited study of TopoLite-KD, a compact COVID-19 CT slice classifier that combines a lightweight CNN, a 134-D persistent-homology descriptor, direct visual–topological gated fusion, and EfficientNet-B0 response distillation. The verified reconstruction achieves 94.62% accuracy, 96.81% sensitivity, and 0.9846 AUROC with 196,773 trainable parameters. The central positive result is the strength of explicit topology; the central negative result is that the complete gated, distilled, Coordinate-Attention

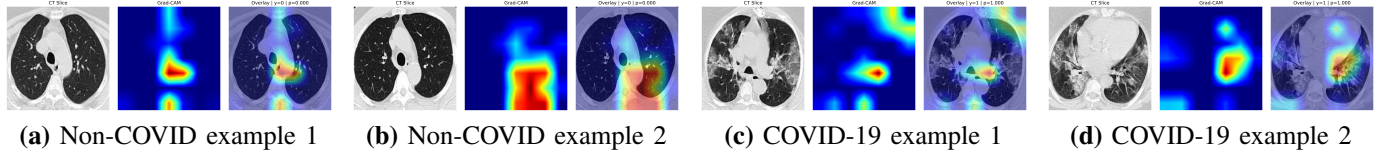


Fig. 5. Qualitative Grad-CAM examples from the verified seed-42 run. Each panel contains the original CT slice, activation map, and overlay. All four cases are correctly classified; no clinical localization claim is made.

TABLE V

PROTOCOL-AWARE COMPARISON AND EVIDENCE STATUS. A DASH MEANS THAT THE INSPECTED SOURCE DID NOT PROVIDE A DIRECTLY COMPARABLE VALUE.

Method / evidence	Evaluation unit	Accuracy	Sensitivity	Specificity	MCC	Interpretation
Kundu <i>et al.</i> fuzzy-integral ensemble [2]	Public paper/repository protocol	98.93%	98.93%	–	–	Published result; four-CNN ensemble; not reproduced under our frozen split.
TopoLite-KD archival record	373 images	96.77%	98.40%	95.11%	0.9359	Author-supplied historical YAML; raw matching artifacts unavailable.
TopoLite-KD audited reconstruction	372 images	94.62%	96.81%	92.39%	0.8932	Fully traceable supplied run; primary verified evidence in this paper.

configuration does not outperform simpler completed ablations. The published 98.93% fuzzy-ensemble result and the archival 96.77% TopoLite-KD record are retained as contextual evidence rather than presented as matched or reproduced comparisons. This framing yields a defensible contribution: a compact and transparent topology-aware pipeline, together with a reproducibility record that clearly distinguishes what was measured from what remains to be completed.

DATA AND CODE AVAILABILITY

The dataset is publicly described by Soares *et al.* [1] and distributed on Kaggle. The evolving source, training code, and reproducibility materials are available in the [TopoLite-KD GitHub repository](#). The assigned reference implementation is available in the fuzzy-integral baseline repository [3]. Before archival publication, the authors should deposit the exact frozen manifest, topology cache signature, teacher and student checkpoints, raw predictions, resolved configurations, and environment lock file for every claimed result.

ETHICS AND INTENDED USE

The supplied dataset is public and this work performs secondary computational analysis. No new patient recruitment or clinical intervention was conducted. The model produces slice-level research predictions and is not validated for diagnosis, triage, treatment, or autonomous clinical decision-making.

REFERENCES

- [1] E. Soares, P. Angelov, S. Biaso, M. H. Froes, and D. K. Abe, “SARS-CoV-2 CT-scan dataset: A large dataset of real patients CT scans for SARS-CoV-2 identification,” *medRxiv*, 2020. doi:10.1101/2020.04.24.20078584.
- [2] R. Kundu, P. K. Singh, S. Mirjalili, and R. Sarkar, “COVID-19 detection from lung CT-scans using a fuzzy integral-based CNN ensemble,” *Computers in Biology and Medicine*, vol. 138, Art. no. 104895, 2021. doi:10.1016/j.compbiomed.2021.104895.
- [3] R. Kundu *et al.*, “Fuzzy-Integral-Covid-Detection: Official implementation,” GitHub repository. [Online]. Available: <https://github.com/Rohit-Kundu/Fuzzy-Integral-Covid-Detection>. Accessed: Jul. 11, 2026.
- [4] M. Roberts *et al.*, “Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans,” *Nature Machine Intelligence*, vol. 3, pp. 199–217, 2021. doi:10.1038/s42256-021-00307-0.
- [5] Y. Singh, C. M. Farrelly, Q. A. Hathaway, T. Leiner, J. M. Jagtap, G. Carlsson, and B. J. Erickson, “Topological data analysis in medical imaging: Current state of the art,” *Insights into Imaging*, vol. 14, Art. no. 58, 2023. doi:10.1186/s13244-023-01413-w.
- [6] S. Iqbal, H. F. Ahmed, T. Qaiser, M. I. Qureshi, and N. Rajpoot, “Classification of COVID-19 via homology of CT-scan,” *arXiv preprint arXiv:2102.10593*, 2021. [Online]. Available: <https://arxiv.org/abs/2102.10593>.
- [7] R. Assaf, A. Rammal, A. Goupil, M. Kacim, and V. Vrabie, “Topological data analysis and machine learning for COVID-19 detection in CT scan lung images,” *BMC Biomedical Engineering*, vol. 7, Art. no. 4, 2025. doi:10.1186/s42490-025-00089-1.
- [8] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” *arXiv preprint arXiv:1503.02531*, 2015. [Online]. Available: <https://arxiv.org/abs/1503.02531>.
- [9] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. 36th Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114. [Online]. Available: <https://proceedings.mlr.press/v97/tan19a.html>.
- [10] X. Li, L. Li, M. Li, P. Yan, T. Feng, H. Luo, Y. Zhao, and S. Yin, “Knowledge distillation and teacher–student learning in medical imaging: Comprehensive overview, pivotal role, and future directions,” *Medical Image Analysis*, vol. 107, Art. no. 103819, 2026. doi:10.1016/j.media.2025.103819.
- [11] Z. Tan, Y. Yu, J. Meng, S. Liu, and W. Li, “Self-supervised learning with self-distillation on COVID-19 medical image classification,” *Computer Methods and Programs in Biomedicine*, vol. 243, Art. no. 107876, 2024. doi:10.1016/j.cmpb.2023.107876.
- [12] Y. Wu and K. He, “Group normalization,” in *Proc. European Conf. Computer Vision (ECCV)*, 2018, pp. 3–19. [Online]. Available: https://openaccess.thecvf.com/content_ECCV_2018/html/Yuxin_Wu_Group_Normalization_ECCV_2018_paper.html.
- [13] Q. Hou, D. Zhou, and J. Feng, “Coordinate attention for efficient mobile network design,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 13713–13722. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2021/html/Hou_Coordinate_Attention_for_Efficient_Mobile_Network_Design_CVPR_2021_paper.html.
- [14] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330. [Online]. Available: <https://proceedings.mlr.press/v70/guo17a.html>.
- [15] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626. [Online]. Available: https://openaccess.thecvf.com/content_ICCV_2017/html/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.html.